Computational dynamics

Computer Laboratory 1:
ParaView (Usage instructions)

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INTRODUCTION

Gmsh → .msh → FEniCS PROJECT → vtk → Paraview

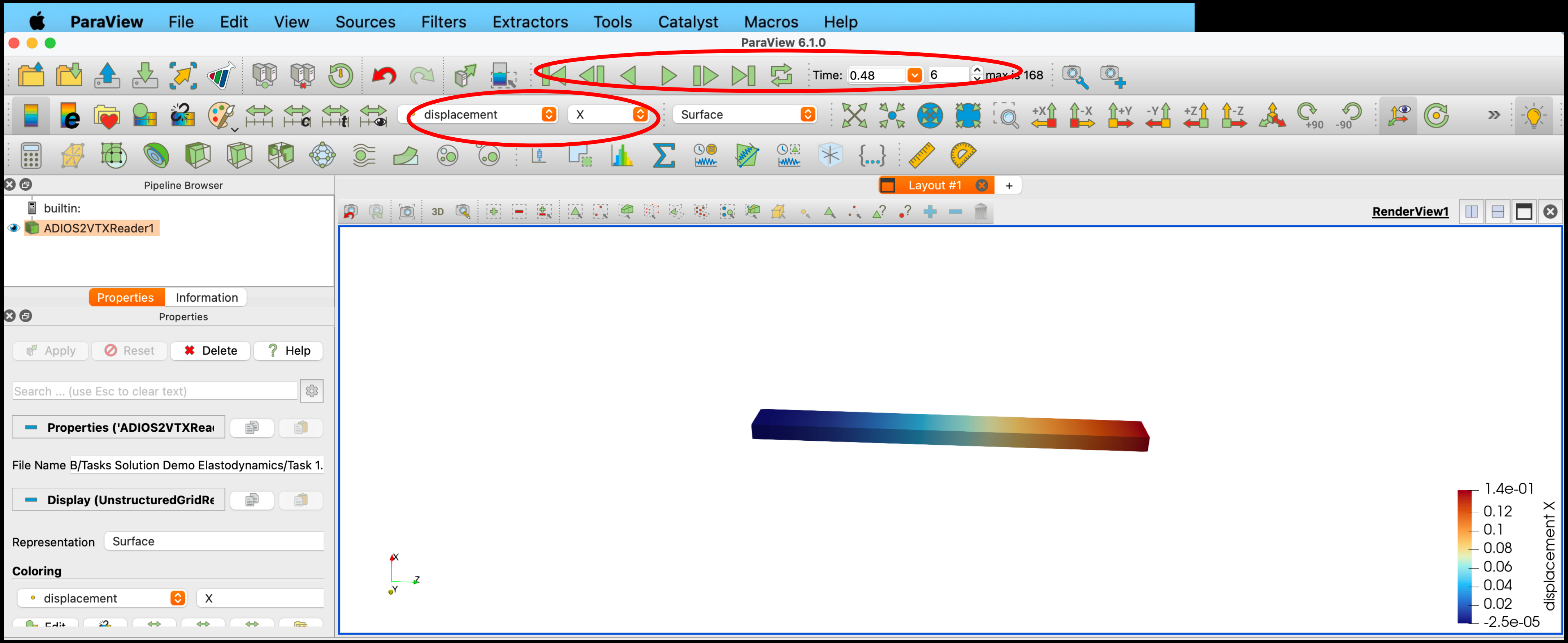
Mesh generator | FE solver | Visualization

- Open-source post-processing visualization engine
- Download: https://www.paraview.org

USING PARAVIEW

Loading the results of a simulation:

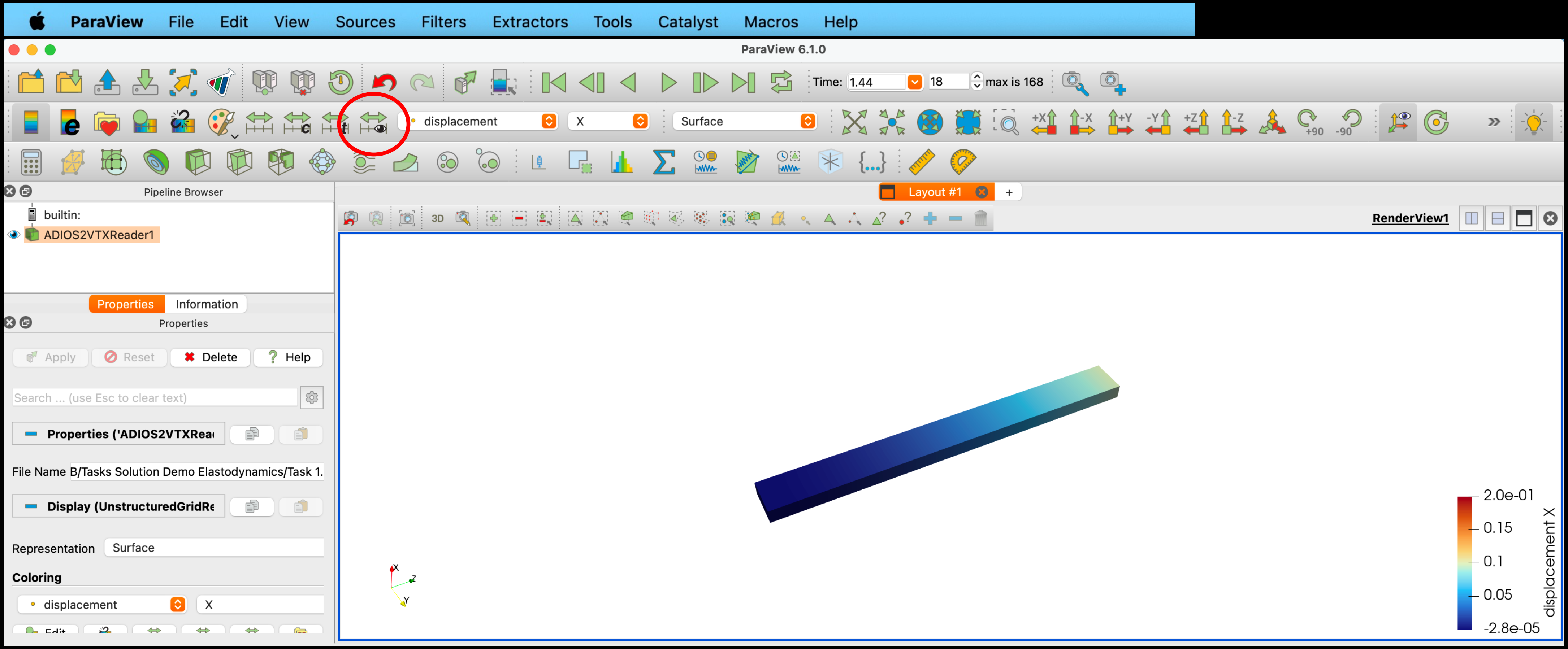
After running (partially/completely) a simulation, open VTK output folder and select ADIOSVTKReader



USING PARAVIEW

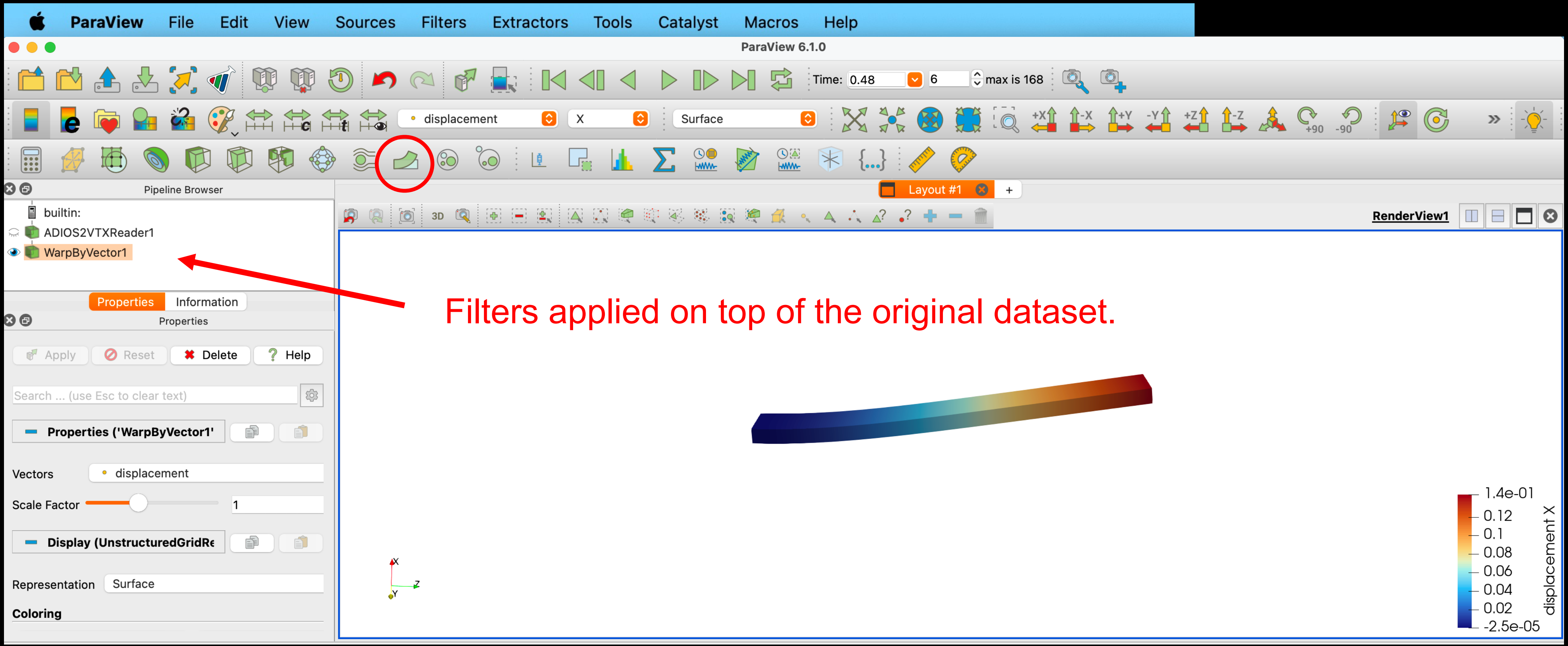
Visualization commands:

Click Rescale to Visible Data Range to adjust the colormap to the data range of the current step.



USING PARAVIEW

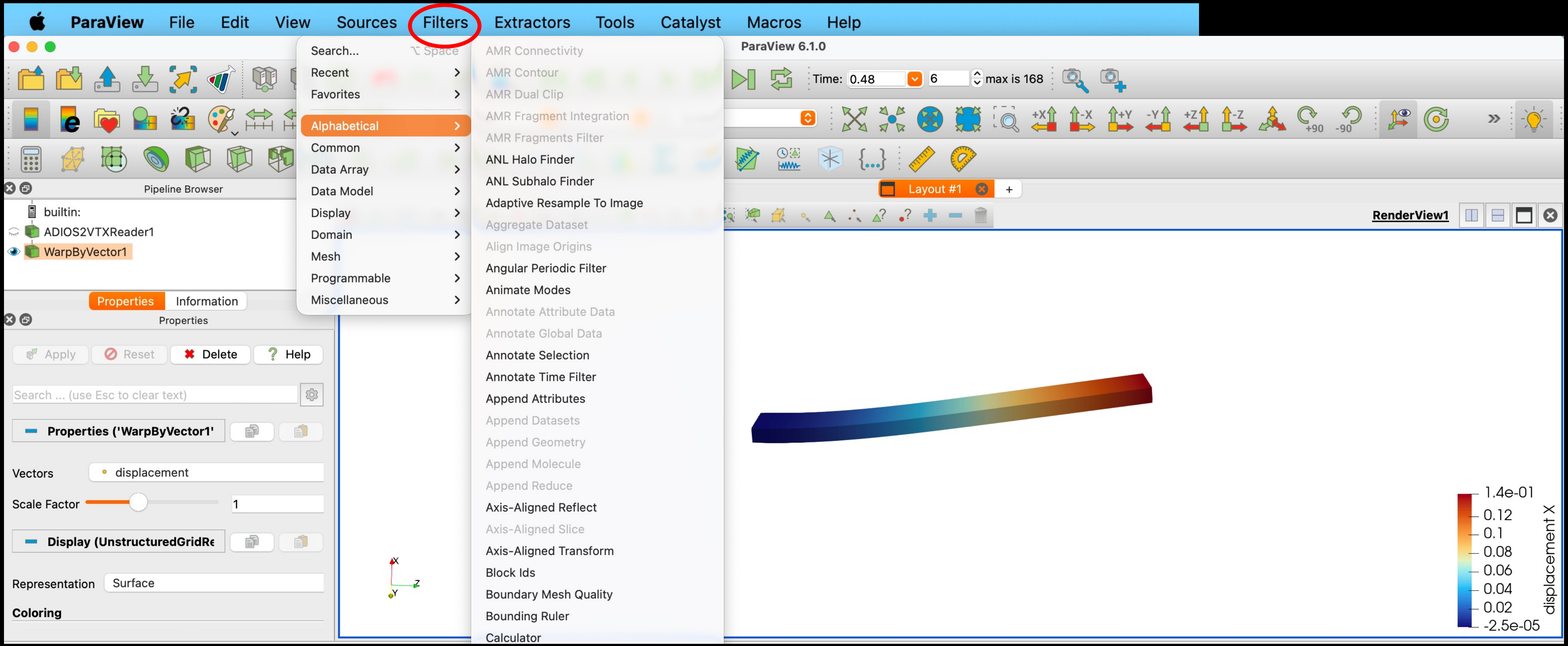
Filters: essential tools for post-processing and visualization; they appear on the Pipeline Browser.
e.g., the *Warp* filter can be used to visualize the deformed configuration.



USING PARAVIEW

Filters: essential tools for post-processing and visualization.

Many other filters in Filters menu



USING PARAVIEW

Hover points: visualize output data directly on the mesh.

Click on *Hover Points On* and move the cursor on the mesh.